Oil Spillage Detector

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Introduction

- Huge quantities of crude oil and refined petroleum products are transported through submarine pipelines.
- They are also transported by ships between production sites.
- It is estimated that approximately 706 million gallons of waste oil enter the ocean every year, which leads to contamination of water and in turn causes a huge risk for aquatic life.
- Our aim is to reduce these effects by precise detection of the location where the pipe was leaked.

Problems Faced

- Proper detection of pipeline leakage

Figure 1.1
Pipeline Leakage

- Oil spill along the coast of Nagapattanam, Tamil Nadu, 28/03/2023

- Area affected by oil spillage

Figure 1.2 Oil spillage

- The bulk carrier Wakashio ran aground on a coral reef near the southern Mauritius in the Indian Ocean, 25/07/2020

Our Solution

- The ocean's average pH is around 8.1, which is basic and in case of oil the pH is around 10 to 11.
- Hence by using a pH sensor we can detect the presence of oil in the water.
- Our project is to develop a boat which has the following features :
  - Solar panel to power up the device
  - A GPS system
  - An arm to hold the pH sensor
- The GPS module is used to navigate the boat to different locations in the ocean where oil pipelines are laid.

Contd....

- Upon reaching different locations, the arm that holds the pH sensor is immersed in water to detect the presence of oil.
- By detecting the presence of oil, it is possible to identify the location where the pipeline is damaged.
- The user gets the information about this location through the GPS system.
- In the case of oil spill from ship the area of spread can be tracked by moving the boat to different areas near the spill and testing the pH value.

Block Diagram

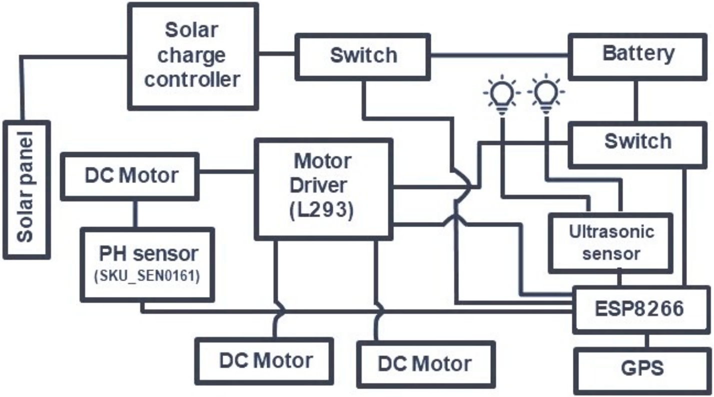

Figure 2.1 Block Diagram of Oil Spillage Detector

ESP8266

- The ESP8266 is a widely used Wi-Fi module in IoT projects.
- It features a microcontroller, Wi-Fi capabilities, and GPIO pins.
- The module provides an affordable solution for wireless connectivity.
- It is commonly used in applications like home automation and sensor networks.
- The ESP8266 is known for its small size and low power consumption.
- It is easy to program and has gained popularity among hobbyists and professionals in the IoT field.

Figure 3.1 ESP8266

L293D Motor Driver

- The L293D is a motor driver IC used to control DC motors and stepper motors.
- It offers bidirectional control, allowing motors to rotate in both forward and reverse directions.
- The L293D can handle a wide range of voltages and is widely used in robotics, automation, and motor control applications.

Figure 3.2 L293D Motor Driver

Johnson Motor

- A Johnson motor with a speed of 600 rpm indicates the rotational speed of its output shaft.
- Additional parameters such as voltage and motor type are needed to provide specific information about a particular Johnson motor model.
- Consulting Johnson Electric or referring to their documentation can provide accurate specifications and details about the motor.

Figure 3.3 Johnson Motor

LIPO Battery

- The LIPO battery has a capacity of 3500mAh and operates at 11.1V.
- It is commonly used in electronics and hobbyist applications such as drones and RC vehicles.
- LIPO batteries provide a balance between capacity and voltage for optimal performance.

Figure 3.4 LIPO Battery

Servo Motor

- The servo motor allows precise control over angular position and velocity.
- Operating at 60 rpm, it delivers smooth and controlled motion for accurate movements.
- With its capabilities, the servo motor finds applications in robotics, automation, and mechanical systems requiring accurate positioning.

Figure 3.5 Servo Motor

Ultrasonic Sensor

- The HC-SR04 is an ultrasonic sensor used for distance measurement in robotics and automation.
- It emits ultrasonic waves and measures the time taken for the waves to bounce back.
- This allows the sensor to calculate the distance between itself and objects in its surroundings.

Figure 3.6 Ultrasonic Sensor

SKU-SEN0161 Ph Sensor

- SKU-SEN0161 is a product identification code used for an electronic sensor.
- Without additional information, it is challenging to provide specific details about the sensor's purpose or specifications.
- For more accurate information, it is recommended to refer to the product documentation or contact the manufacturer directly.

Figure 3.7 SKU-SEN0161 Ph Sensor

JQC3F Relay

- The JQC-3F relay is an electromagnetic relay commonly used for switching applications.
- It is compact in size and versatile, capable of handling moderate current and voltage levels.
- With multiple contacts, including normally open (NO) and normally closed (NC) options, it offers flexibility for various electrical control purposes in industrial, home automation, and automotive systems.

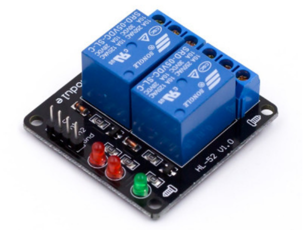

Figure 3.8 JQC3F Relay

Solar Panel

- A 5-watt solar panel has a power output of 5 watts, making it suitable for small-scale applications.
- It is commonly used for charging small electronic devices and powering small LED lights.
- Due to its limited power output, it is not typically used as the primary power source for larger energy-consuming devices or systems.

Figure 3.9 Solar Panel

CN3791 Solar charger controller

- CN3791 solar charger: Compact and efficient device.
- Utilizes solar power to charge batteries.
- Includes MPPT algorithm for optimized energy conversion.
- Equipped with protection features for safe charging.
- Suitable for portable solar power systems and off-grid applications

Figure 3.10 CN3791 Solar charger controller

Arduino IDE

- Arduino IDE offers a user-friendly interface and simplifies code compilation and uploading to Arduino boards.
- It provides an extensive library of pre-written code and a serial monitor for easy hardware interaction.
- Arduino IDE benefits from an open-source community, offering a wealth of resources, tutorials, and collaborative support.

Figure 4.1 Arduino IDE

Blynk IoT

- Blynk offers a user-friendly interface for creating IoT applications.
- It provides a mobile app and cloud-based infrastructure for seamless device connectivity and control.
- Blynk is accessible to beginners and experts, enabling IoT project development without extensive programming knowledge.

Figure 4.2 Blynk IoT

Structure of boat

- Material used: Thermocol
- Reason: Thermocol has lesser density than the density of liquid water and hence keeps it afloat.
- 3 layers of thermocol constitutes the body.
- A thin wooden plank is used as the base of the structure to support the thermocol.
- For the movement two paddles are used, which is made up of two normal wheels modified by attaching thin rectangular steel sheets along its outer surface.

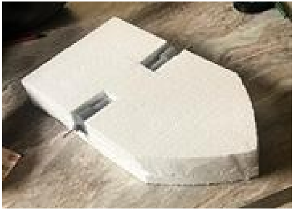

Figure 5.1.1 Base structure

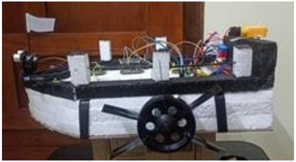

Figure 5.1.2 Side view

Navigation

- For the navigation purpose, the ESP board is interfaced with the Blynk IoT app.
- Digital pins of ESP board is controlled with the help of Blynk IoT app and it is connected to an external motor driver.
- For the forward movement, the clockwise rotation of the wheels is done by giving input as 1 and 0.
- Likewise for the backward movement the input is given 0 and 1.

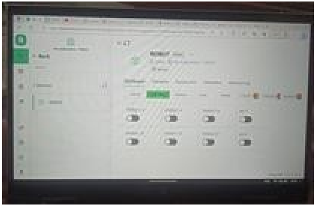

Figure 5.2.1 Blynk IoT

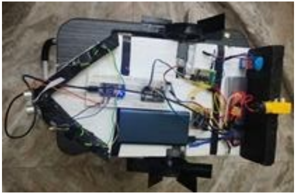

Figure 5.2.2 Ariel view

Functionalities - pH Sensor Holder

- pH of the liquid is measured by using a pH sensor.
- An arm is used to hold the pH sensor.
- The arm is connected to a servo motor which is in turn connected to a motor driver.

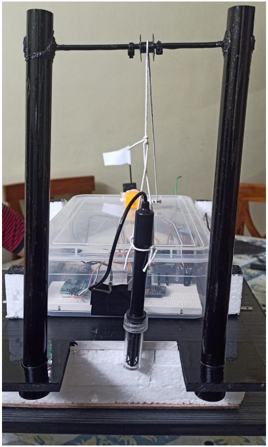

Figure 5.3 pH sensor holder

Functionalities - Ultrasonic Sensor

- An ultrasonic sensor is used in order to prevent the structure from colliding with any obstacles present before.
- The sensor is placed at the front end of the structure.

Figure 5.4 ultrasonic sensor

Functionalities - pH Sensor

- The pH sensor is used to determine the pH value of liquid
- It helps to identify the presence of oil

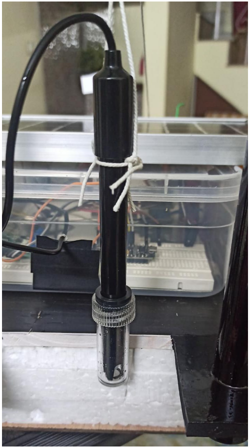

Figure 5.5 pH sensor

Functionalities -Gps

- It is used for navigation purpose
- It tracks and monitors navigation
- It gives the location at which the pipeline is leaked

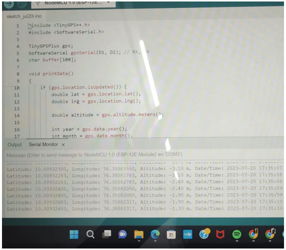

Figure 5.6 GPS readings

Functionalities -Solar panel

- Solar energy is used to power up the boat.
- solar panel=5W
- solar charge controller=2A
- solar panel wattage=(battery capacity * battery voltage)
- solar charge controller size =(power produced by solar panel /voltage of batter)

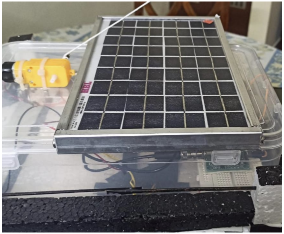

Figure 5.7 Solar panel

Work Division

- BOAT NAVIGATION
  Dana johnson
  Albert Johnson
- PH SENSING
  Aaggi Rose Thomas
  Baasil Samaan AA
- REPORT
  Aaggi Rose Thomas
  Dana johnson
- FINAL PPT
  Albert Johnson
  Baasil Samaan AA

Cost Estimation

- Johnson Motor(2) –1400 Rs/-
- Servo Motor –300 Rs/-
- ESP8266 –575 Rs/-
- Motor Driver –300 Rs/-
- LIPO Battery –3500 Rs/-
- PH Sensor –2500 Rs/-
- GPS –500 Rs/-
- Solar Panel - 500 Rs/-
- Solar Charge Controller- 350 Rs/-
- Additional Cost –2000 Rs/-
- TOTAL=10,000 Rs/-

Reference

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- Chen, Y., Li, X., & Wang, Q. (2022). pH Sensors for Water Quality Monitoring: A Review. IEEE Sensors Journal, 20(5), 2345-2356.
- The Hindu, "Crack in underwater pipeline causes oil spill in sea off Nagapattanam in Tamil Nadu" - by C.Jaisankar (March 03/2023, 06:23pm).

Thank you!